Louis Whittome

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PERSONAL PROFILE

Final-year BEng Electrical and Electronic Engineering student and scholarship athlete at the University of Warwick, with First-Class grades to date. Passionate about developing innovative hardware and energy systems, with hands-on experience spanning PCB assembly and embedded software in industry through to MEMS gas-sensor research. Motivated to apply technical skills to real-world challenges in electronics, renewable energy, and emerging technologies.

EDUCATION

BEng Electrical and Electronic Engineering — University of Warwick *(2023 – 2026, expected June 2026)*

First-Class grades to date. Key modules: Analogue Electronics Design, Power Electronics, Embedded Systems, Computer Architecture & Systems, Semiconductor Materials & Devices, Control Engineering, Electromechanical Systems, Modelling Simulation & Computation.

A-Levels — Colyton Grammar School, Devon: Mathematics (A*), Further Mathematics (A*), Physics (A*), Chemistry (A*)

GCSEs — The Woodroffe School, Lyme Regis: eleven GCSEs including five 9s and five 8s

FINAL-YEAR PROJECT

MEMS Thermal Conductivity Gas Sensing for Methane & Hydrogen Detection *(2025 – 2026)* — supervised by Prof. Julian Gardner, in partnership with Flusso Ltd

Evaluated a MEMS thermal conductivity sensor (Flusso FLS310 IC) for detecting trace methane and hydrogen mixtures in air (0–1.2% Vol), relevant to industrial safety and the UK hydrogen grid. Designed and built the experimental platform: custom PCB with differential amplification, controlled gas-flow rig, and Arduino-based data acquisition over I^2C. Characterised response across humidity and temperature profiles and developed a multi-variable regression model compensating for environmental factors, achieving $R^2 = 0.969$ (RMSE 0.165 Ω) — a successful proof of concept for low-cost gas detection.

PROFESSIONAL EXPERIENCE

Assistant Electronics Engineer — EDM Solutions Ltd, Manchester *(Aug 2024)*

Contributed to full production of flight simulator handsets for Air India: soldered and assembled PCBs interfaced with Raspberry Pi systems, calibrated embedded code, and performed functionality testing. Developed insight into hardware–software integration and professional design workflows.

Assistant Engineer — PDQ Precision Ltd, Yeovil *(Sept 2022)*

Assisted in precision component manufacturing using CNC lathe and milling machines. Gained experience in CAD-based part design, tolerancing, and design for manufacture.

Water Sports Instructor — Mark Warner, Rhodes, Greece *(Jun – Sept 2023)*

Led RYA-certified sailing and windsurfing courses for children and adults, managing safety-critical activities and strengthening communication, leadership, and public-speaking skills.

Kitchen Hand — The Alpinist, Niseko, Japan *(Dec 2022 – Apr 2023)*

Worked in a high-end restaurant serving up to 100 three-course meals nightly, developing resilience, teamwork, and efficiency under pressure.

TECHNICAL SKILLS

Programming: C, Python, MATLAB, Verilog, Assembly, Arduino IDE

Hardware: Microcontrollers, PCB design fundamentals, soldering & assembly, electronic instrumentation, embedded systems

Design & tools: Fusion 360 (assemblies, motion and stress simulation), CNC machining exposure, Microsoft Excel/Word/PowerPoint

ADDITIONAL

Sports scholarship athlete at the University of Warwick; RYA-qualified sailing and windsurfing instructor; competed nationally in trampolining and at county level in badminton and swimming. Full clean UK driving licence. References available on request.